

Next Generation Data Centers and Energy Management Systems

Lecture 2A: Psychrometrics and Heat Transfer

Prof. Thomas M. Schutzius

Department of Mechanical Engineering

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Learning Plan

- Introduction to Data Center Systems and Thermodynamics
- **Psychrometrics and Heat Transfer**
- Performance Metrics and Standards
- Applied Computational Fluid Dynamics (CFD)
- Liquid Cooling & Advanced Heat Transfer
- Chip Cooling: Heat Conduction and Thermal Interface Materials
- Power Delivery and Infrastructure (Energy sources: Geothermal, Solar, Nuclear, etc.)
- Overview of Simulation Field; the Concept of a Digital Twin
- Machine Learning and Optimization Tools
- Modeling and Simulation of an Energy Management System
- Control Systems and Optimization
- Spacecraft Thermal Control
- Final Project Work, Peer Feedback, and Mentor Check-in
- Symposium

Learning Objectives

- Define different data center components, layouts, and energy standards.
- Analyze a data center cooling infrastructure using thermodynamic and psychrometric principles.

Typical layout of a data center facility

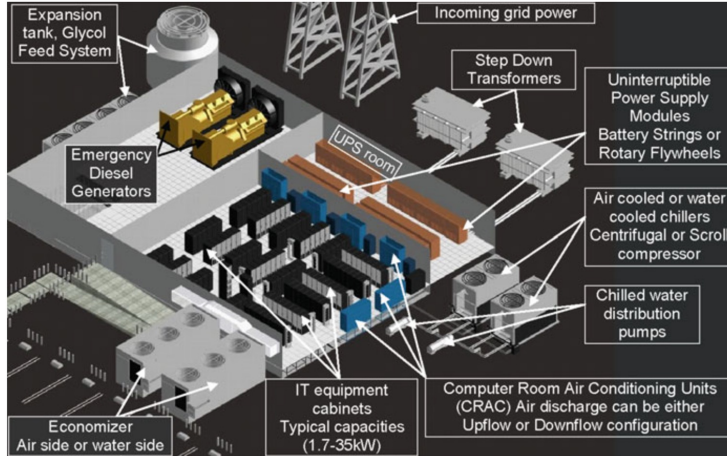


Figure 1: Reproduced from Figure 1.1 in Ref. [2].

Energy flow in a data center

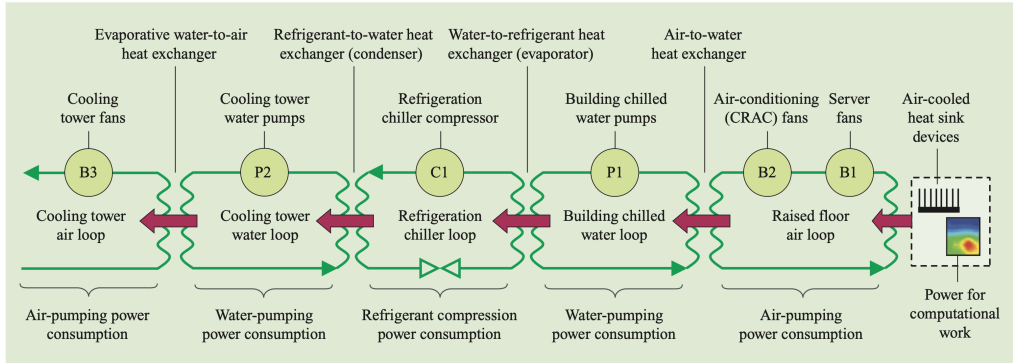


Figure 2: Reproduced from Figure 5 in Ref. [4].

Cooling infrastructure

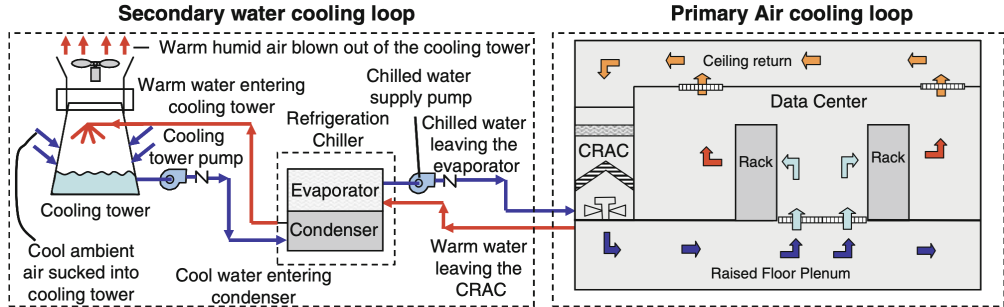


Figure 3: Reproduced from Figure 2.4 in Ref. [2].

Psychrometrics of the rack

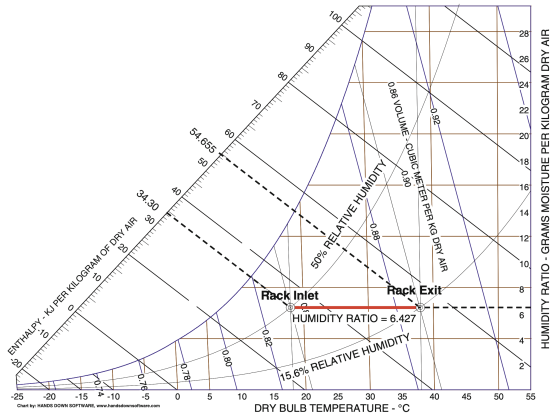


Figure 4: Reproduced from Figure 2.44 in Ref. [2].

Energy analysis of heat transfer from a cabinet

- In addition to maintaining the inlet air temperatures, data centers also need to adhere to the humidity levels specified by ASHRAE guidelines.
- ASHRAE: American Society of Heating, Refrigerating and Air-Conditioning Engineers
- The relevant technical committee is: TC 9.9 (Mission Critical Facilities, Data Centers, Technology Spaces and Electronic Equipment): The flagship committee for data centers. TC 9.9 sets global industry standards for thermal guidelines.
- Standard 90.1-2025 – Energy Standard for Data Centers: [Link](#). You need to register in order to read. You can also become a student member of ASHRAE. There is a Golden Gate chapter.

Hot-aisle/cold-aisle layout

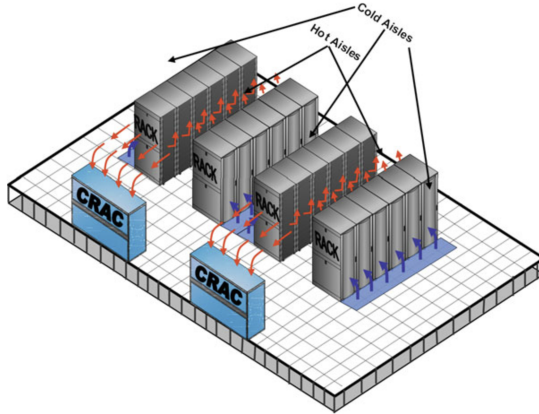


Figure 5: Reproduced from Figure 1.15 in Ref. [2].

The ideal vapor-compression refrigeration cycle

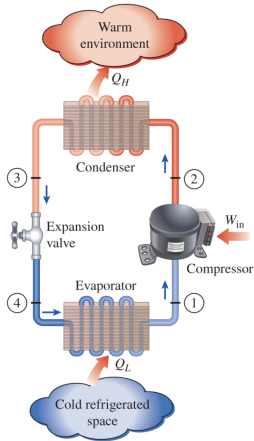


Figure 6: Reproduced from Figure 11.3 in Ref. [1].

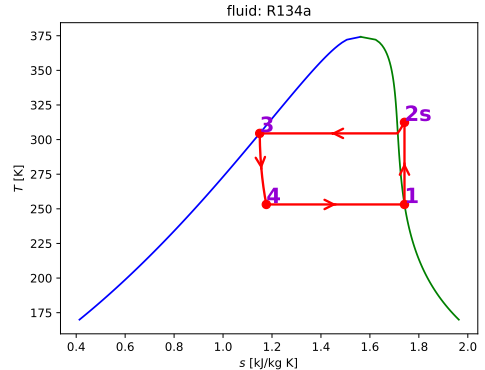


Figure 7: Generated from [here](#).

Economizers

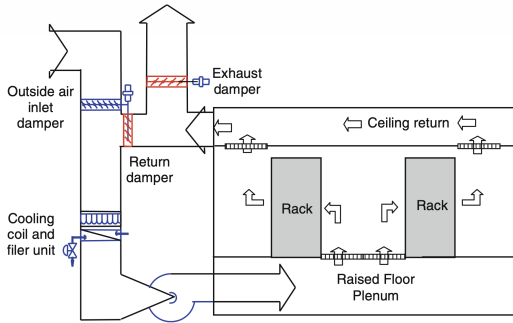


Figure 8: Reproduced from Figure 2.51 in Ref. [2].

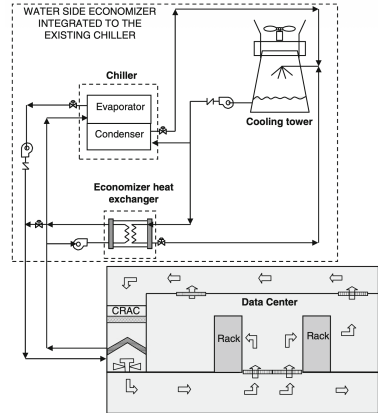


Figure 9: Reproduced from Figure 2.53 in Ref. [2].

Water-side Economizer

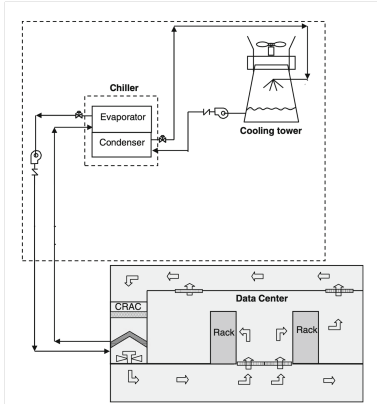


Figure 10: No Economizer.

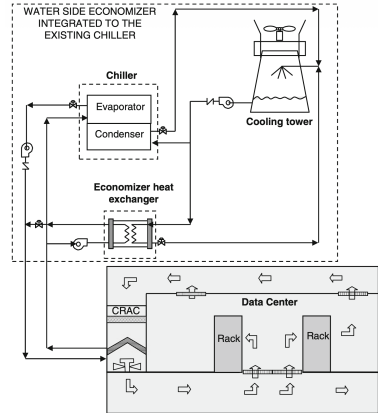


Figure 11: Water-side Economizer (Figure 2.53 in Ref. [2]).

High-power chip heat sink

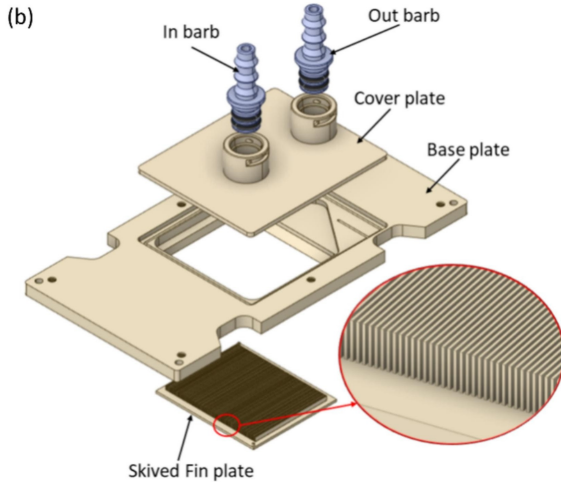


Figure 12: Exploded view of a skived fin cold plate (Figure 7b in Ref. [3]).

Liquid-cooled rack



Figure 13: Reproduced from [Cool IT](#).

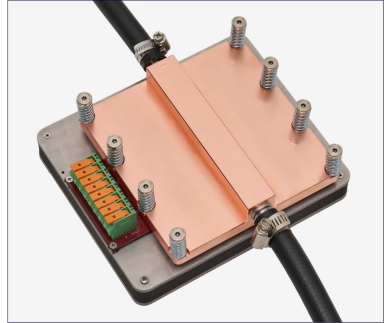


Figure 14: 15 kW cold plate mounted to a thermal test vehicle. Reproduced from [Cool IT](#).

Cooling infrastructure: Psychrometric principles

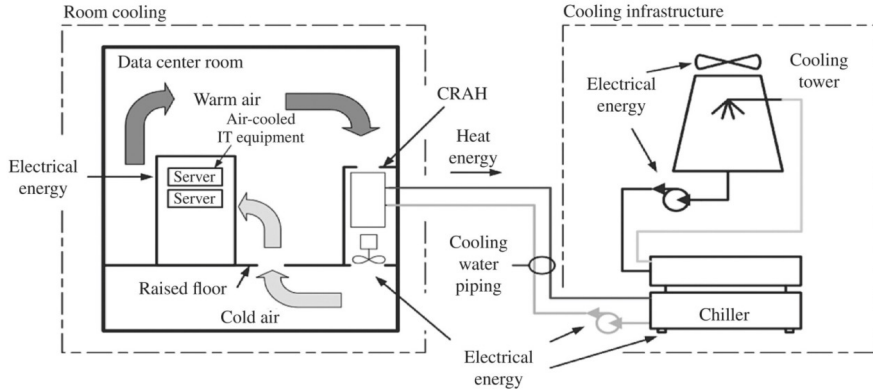


Figure 15: H. Geng. Data Center Handbook. 2nd Edition, Wiley, 2021. Chapter 14. Example: [psychrometricPrinciples.ipynb](#)

Cooling infrastructure: Cooling tower

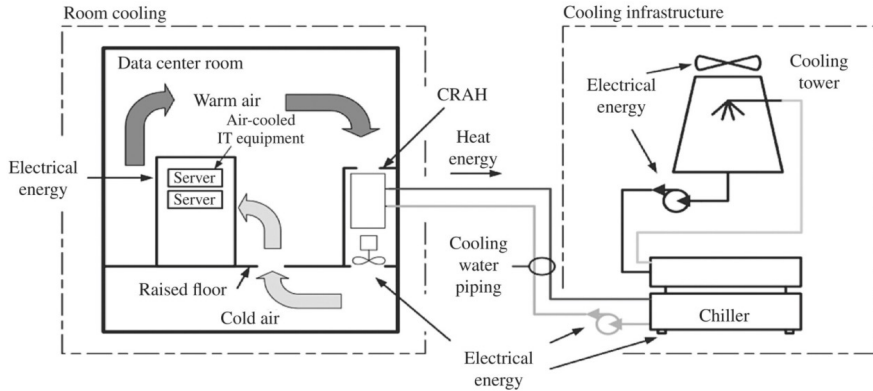


Figure 16: H. Geng. Data Center Handbook. 2nd Edition, Wiley, 2021. Chapter 14. Example: [coolingTower.ipynb](#)

References I

- [1] Yunus A Cengel, Michael A Boles, and Kanoğlu.
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- [2] Yogendra Joshi and Pramod Kumar, editors.
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- [3] Sangram Kumar Samal, Hsien-Chun Chang, Yogesh Fulpagare, and Chi-Chuan Wang.
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References II

- [4] R. Schmidt and M. Iyengar.
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